OWNML MACHINE LEARNING CANVAS

Designed for: Designed by: Date: Iteration: .

PREDICTION TASK

What is the type of task?
Which entity are predictions made on?
What are the possible outcomes to predict?
When are outcomes observed?

DECISIONS

How are predictions turned into actionable recommendations or decisions for the end-user? (Mention parameters of the process / application for this.)

VALUE PROPOSITION

Who is the end beneficiary, and what specific pain points are addressed?
How will the ML solution integrate with their workflow, and through which user interfaces?

DATA COLLECTION

How is the initial set of entities and outcomes sourced (e.g., database extracts, API pulls, manual labeling)?
What strategies are in place to update data continuously while controlling cost and maintaining freshness?

DATA SOURCES

Where can we get data on entities and observed outcomes? (Mention internal and external database tables or API methods.)

IMPACT SIMULATION

What are the cost/gain values for (in)correct decisions?
Which data is used to simulate pre-deployment impact?
What are the criteria for deployment?
Are there fairness constraints?

MAKING PREDICTIONS

Are predictions made in batch or in real time?
How frequently?
How much time is available for this (including featurization and decisions)?
Which computational resources are used?

BUILDING MODELS

How many models are needed in production?
When should they be updated?
How much time is available for this (including featurization and analysis)?
Which computation resources are used?

FEATURES

What representations are used for entities at prediction time?
What aggregations or transformations are applied to raw data sources?

MONITORING

Which metrics and KPIs are used to track the ML solution's impact once deployed, both for end-users and for the business?
How often should they be reviewed?

From framing to funding Machine Learning

The **Machine Learning Canvas** is typically the first step. Teams use it to collaboratively frame a machine learning use case, surface assumptions early, and align on what a first experiment should actually prove — before models are built and resources are committed.

In practice, filling the Canvas often raises broader questions, such as:

- Which ML use cases should we prioritize across teams?
- How do we compare value, complexity, and confidence?
- How do we decide whether a pilot is worth scaling?
- How do we discuss ML progress in business terms, not just technical metrics?

These questions are less about modeling — and more about **discipline**.

Beyond the Canvas

OWNML builds on the Machine Learning Canvas to help organizations bring discipline to enterprise machine learning.

It extends the Canvas with shared artifacts and practices that support:

- portfolio-level prioritization of ML use cases
- explicit assumptions and impact simulation
- scorecards and one-pagers leadership understands
- consistent decision-making across teams

The goal is not to accelerate experimentation at all costs, but to help teams **decide what to build, fund, and scale — with confidence**.

How teams typically use the Canvas

Teams often use the Canvas in a short working session (60–120 minutes) to:

- frame a concrete ML use case
- clarify who would act on the prediction and why
- identify assumptions that must hold for value to exist

This shared framing is what makes subsequent technical work meaningful — and defensible.

Learn more

If this way of working resonates, you can learn more about the broader OWNML approach at:

https://www.ownml.co/

Referencing the Machine Learning Canvas

If you need to reference the Canvas formally (e.g. in documentation or publications), you may use:

```
@misc{mlc,
  title  = "Machine Learning Canvas",
  author = "Louis Dorard (OWNML)",
  url    = "https://www.ownml.co/",
  year   = "2015"
}
```